MIL-T-5383(USAF)
26 June 1950
Superseding
USAF No. 32498-B
25 July 1949

MILITARY SPECIFICATION

TRANSFORMER, CURRENT, TYPE D-1

1. CLASSIFICATION

1.1 Type.- This specification covers one type of aircraft instrument current transformer, designated Type B-1.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following specifications, bulletin, and other publications, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications:

2.1.1.1 Military:

MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-P-6064	Packaging of Lightweight Aircraft Accessories

2.1.1.2 Air Force-Navy Aeronautical:

AN-M-13	Marking, Labeling, and Tagging; Aircraft Aeronautical Equipment, Accessories, Parts, Materials and Supplies

2.1.1.3 U. S. Air Force:

32497	Ammeter; AC, 0-250 Amps, Type H-1
32500	Watt-Varmeter; Aircraft, 400 Cycle AC, Type A-1
41065	Equipment; General Specification For Environmental Test of

2.1.2 Specification Bulletin:

2.1.2.1 Air Force-Navy Aeronautical:

143	Specifications and Standards; Use of

2.1.3 Other Publications:

C57.10-1948	American Standards Association Standard

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3. REQUIREMENTS

3.1 Material:

3.1.1 Specifications and standards.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143, except as provided in the following paragraph:

3.1.1.1 Standard parts.- AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they develop suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.1.2 Protective treatment.- When materials are used in the construction of the transformer that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.2 Design and construction.- The transformer shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.2.1 Rating.- The current transformer shall be rated at 250:2 amp. It shall have a current ratio of 125:1 with the primary conductor passing once through the aperture in the transformer secondary.

3.2.2 Burden.- The transformer shall be designed for maximum accuracy at a line power factor of 0.75 and a normal volt ampere burden consisting of an ammeter, watt-varmeter, and 100 circuit feet of No. 18 copper wire as shown on Figure 1. The total burden impedance shall be 1.48 + j 2.09 ohms ±10 percent at 400 cycles.

3.2.2.1 The phase angle error shall not exceed 2 degrees, and the ratio error shall not exceed 5 percent under the following operating conditions:

3.2.2.1.1 With burden impedance not greater than 13.3 + j 7.4 ohms and primary current between 25 and 111 amp.

3.2.2.1.2 With burden impedance not greater than 5.9 + j 3.3 ohms and primary current between 111 and 250 amp.

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DIAGRAM OF CONNECTIONS OF 400-CYCLE AIRCRAFT INSTRUMENTS AND CURRENT TRANSFORMERS

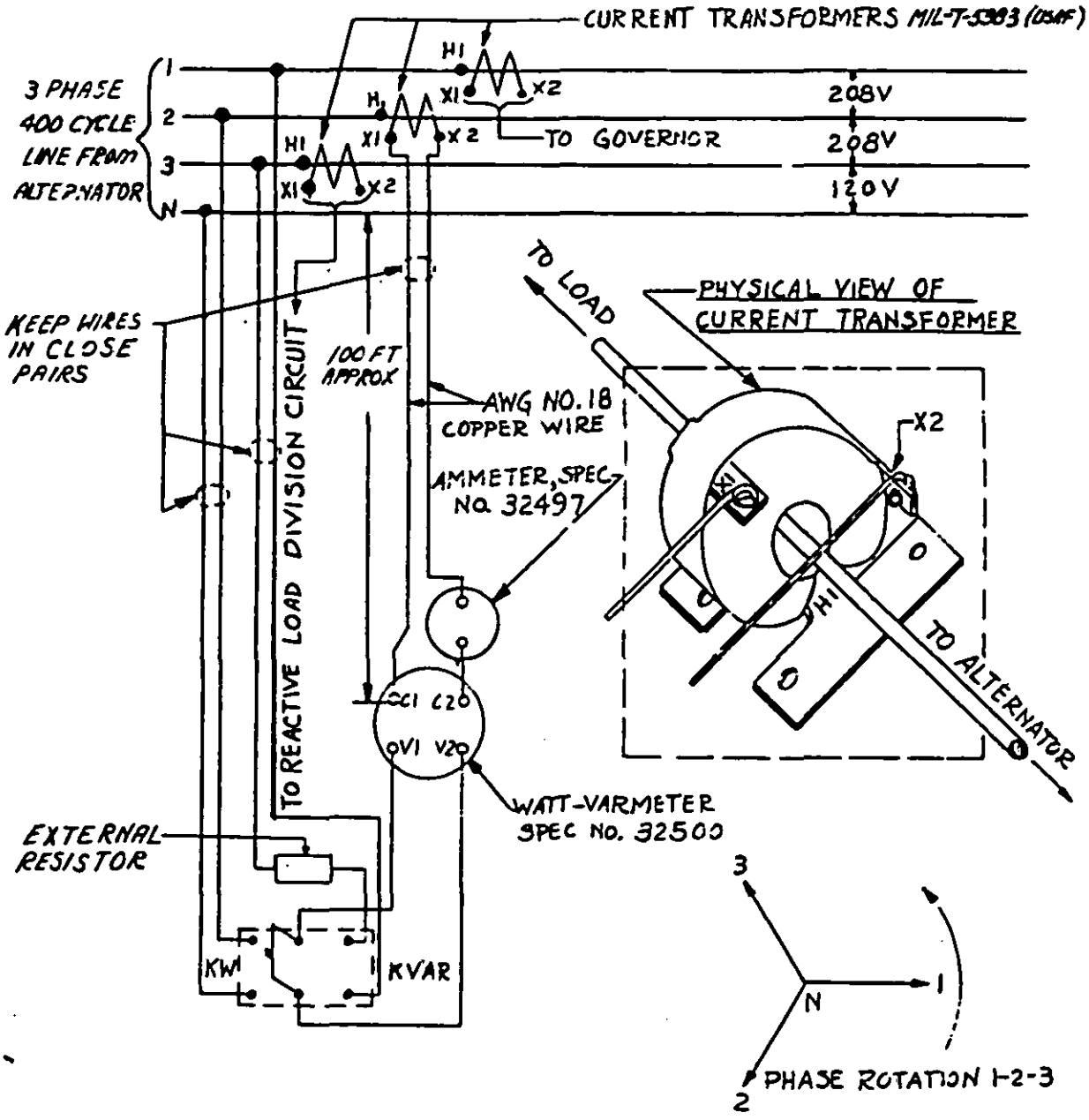


FIGURE 1

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3.2.3 Terminals.- The terminals of the secondary winding of the current transformer shall be copper strips.

3.2.4 Polarity.- The polarity of the transformer shall be designated by the symbols M1 and X1 in accordance with American Standards Association Standard C57.10-1948.

3.2.5 Frequency influence.- The frequency influence on the transformer shall be zero percent at 400 cycles and shall not exceed 0.5 percent for any frequency between the limits of 350 to 450 cycles.

3.2.6 Temperature rise.- The temperature of the windings shall not exceed 50°C for normal burden in an ambient temperature of +25°C.

3.2.7 Accuracy.- The accuracy of the transformer shall be defined in terms of percent of actual change in instrument reading due to the transformer errors.

3.2.7.1 The accuracy of the transformer, with normal burden at +25°C ambient temperature, shall not introduce errors in the ammeter, watt-varmeter (measuring watts), and watt-varmeter (measuring vars when suitable external resistance is used) readings of more than ±1 percent.

3.2.8 Insulation.- The current transformer shall be designed to withstand a potential of 2500V rms ac, commercial frequency, at sea level, applied between the terminals and mounting base for a period of 1 minute.

3.3 Performance.- The transformer shall be capable of operating under the following conditions:

3.3.1 Temperature.- At ambient temperatures from -65 to +70°C.

3.3.2 Humidity.- Relative humidity ranging up to 100 percent.

3.3.3 Vibration.- Vibration consisting of simple harmonic motion.

3.3.4 Fungus.- Fungus growth in tropical climates.

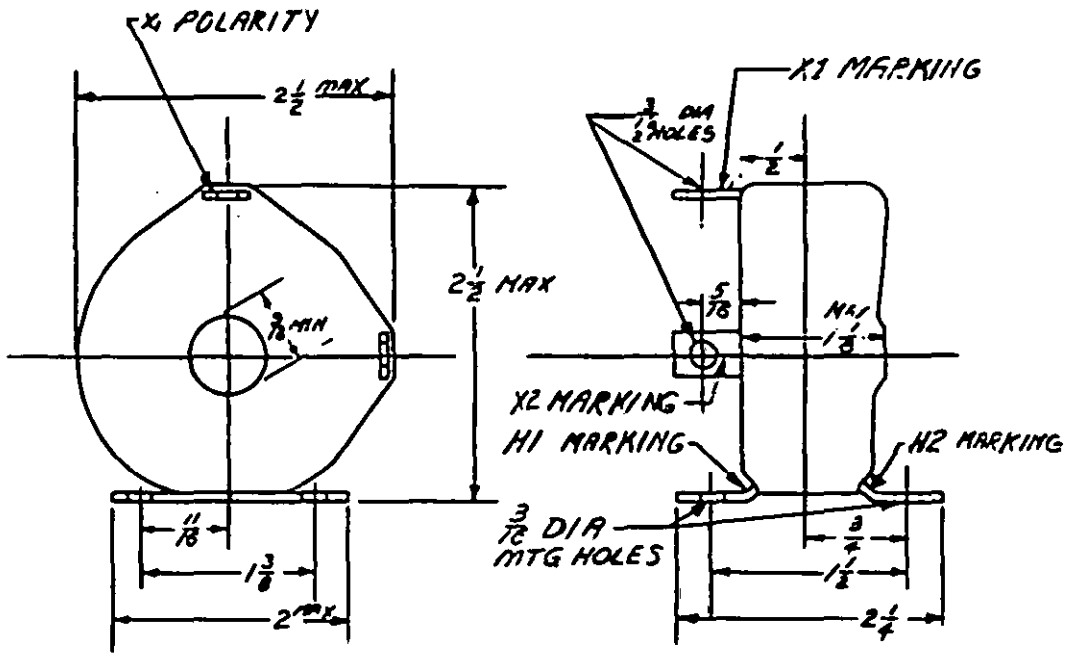
3.3.5 Salt Spray.- Atmosphere containing salt-laden moisture.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Dimensions.- The transformer shall be of the window type and shall conform to the dimensions shown on Figure 2.

3.6 Weight.- The weight of the transformer shall not exceed 0.6 pound.

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INSTRUMENT CURRENT TRANSFORMER

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES:
—ANGLES—
—DECIMALS—
—FRACTIONS— ± 1/64

FIGURE 2

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3.7 Markings.- The transformer shall be permanently and legibly marked as shown on Figure 2. All markings shall be durable to withstand usage encountered in service.

3.7.1 Nameplate.- A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the transformer:

Transformer, Current
Type D-1
Specification MIL-T-5383
Mfr's. Part No. _____
Mfr's. Serial No. _____
Order No. _____
Stock No. _____
(Mfr's. Name or Trademark)
U S Property
(FOR 200/115V 3 PH 400 CPS 4 WIRE SYS)

3.7.1.1 Manufacturer's part number.- The manufacturer's part number marked in the space provided in the nameplate shall be in accordance with Specification AN-M-15.

3.7.1.2 Order number.- For Air Force procurement, where both the purchase order number, such as "AF 33(038) 50-4710" and the contract number, such as "AF 33(038)-1565", are given on an order or contract, the number placed on the nameplate in the place designated "Order Number" shall be the number corresponding to the Air Force's purchase order number, for example: "50-4710". In the absence of the Air Force purchase order number, and the contract number, such as "AF 33(038)-1565", is given, the letters "AF" and the last digits of the number shall be placed on the nameplate, for example: "AF-1565".

3.7.1.3 Stock number.- The stock number shall be as specified by the procuring agency. The stock number shall include the name or standard abbreviations of the name of the service having jurisdiction of the equipment.

3.8 Workmanship:

3.8.1 General.- The transformer, including all parts and accessories shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.8.2 Dimensions.- Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transformer, they shall be held or limited accordingly.

3.8.3 Screw assemblies.- Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

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3.8.4 Cleaning.- The transformer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of transformers shall be classified as follows:

4.1.1 Qualification tests.- Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

4.1.2 Inspection tests.- Inspection tests are those tests accomplished on transformers submitted for acceptance under contract.

4.2 Qualification test:

4.2.1 Prior qualification.- Unless otherwise specified by the procuring agency, transformers which have not previously passed a qualification test, or which have passed the qualification test and have been modified in any manner, shall satisfactorily pass a qualification test prior to the acceptance of any transformer.

4.2.2 Qualification test samples.- Qualification test samples shall consist of 2 transformers. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter.

4.2.3 The qualification test shall consist of all of the tests specified herein, subject to the test conditions specified in 4.4 and shall be conducted in the following order:

Current Transformer No. 1	Current Transformer No. 2
Examination of Product	Examination of Product
Accuracy	Accuracy
Temperature Rise	Insulation
Insulation	Mildew Resistance
Vibration	
High Temperature	
Low Temperature	
Humidity	
Salt Spray	

4.2.3.1 Environmental.- The transformer shall be subjected to the following environmental tests conducted in accordance with Specification No. 41065. Upon completion of each test, the transformer shall show no signs of damage and, unless otherwise specified, shall withstand the insulation test, using a potential of 1000V in lieu of 2500V.

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4.2.3.1.1 Vibration.- Method 63.

4.2.3.1.2 High temperature.- Method 12. The insulation test shall not be conducted.

4.2.3.1.3 Low temperature.- Method 21. The insulation test shall not be conducted.

4.2.3.1.4 Humidity.- Method 31.

4.2.3.1.5 Salt spray.- Method 51 for a period of 50 hours.

4.2.3.1.6 Mildew resistance.- Method 71.

4.3 Inspection tests:

4.3.1 Inspection samples:

4.3.1.1 One transformer shall be selected at random from each lot of 250 or fraction thereof on the order and subjected to the sampling tests. These tests shall be in addition to the individual tests specified herein.

4.3.1.2 When tests are specified on a quantity of transformers that are selected as representative of a certain lot, and 1 or more of this number fails to meet the requirements, additional transformers of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the transformer.

4.3.2 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be accomplished on articles to be supplied on the contract or order. The record or report of inspection and tests shall be signed and approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.3.3 Inspection of the transformer shall consist of the tests specified under inspection test methods, subject to the test conditions specified herein.

4.4 Test conditions:

4.4.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure

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(approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.5 Inspection test methods:

4.5.1 Individual test.- Each transformer shall be subjected to the following test:

4.5.1.1 Examination of product.- The transformer shall be inspected to determine compliance with this specification with respect to workmanship, construction, weight, dimensions, and markings.

4.5.2 Sampling tests:

4.5.2.1 Accuracy:

4.5.2.1.1 The accuracy of the transformer shall be measured at 350, 400, and 450 cycles with burden as shown on Figure 1, or the equivalent thereof, and line power factor of 0.75. The transformer shall introduce errors in the meter readings of not more than ±1 percent of full scale value at 400 cycles, and of not more than ±1.5 percent of full scale value to 350 and 450 cycles.

4.5.2.1.2 The phase angle error and the ratio error of the transformer shall be measured under the following operating conditions:

With burden impedance of 13.3 + j 7.4 ohms and primary currents of 25 and 111 amps.

With burden impedance of 5.9 + j 3.3 ohms and primary currents of 111 and 250 amps.

Under the above operating conditions, the phase angle error shall not exceed 2 degrees and the ratio error shall not exceed 5 percent.

4.5.2.2 Temperature rise.- The temperature rise of the current transformer shall be measured at 25°C ambient with burden impedance of 5.9 + j 3.3 ohms and a primary current of 250 amp. Under these conditions the temperature of the windings shall not exceed 50°C.

4.5.2.3 Insulation.- A potential of 2500V rms ac at commercial frequency shall be applied between terminals and metal mounting base for a period of 1 minute. There shall be no breakdown of insulation as a result of this test.

4.6 Retest.- Rejected transformers shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

4.7 All parts, specimens, or assemblies destroyed in making tests required by this specification and/or drawings, to determine compliance with the specification and/or drawings, shall be in addition to the quantity specified

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in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 General.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 The transformer shall be packaged and packed for shipment in accordance with Group VII of Specification MIL-P-6064 in a Type II, Class I container.

5.3 Marking.- Marking shall be in accordance with the requirements of Specification MIL-P-6064 and as specified herein.

5.3.1 Interior packages.- Each interior package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

Transformer, Current, Rating 250:2 Amp 400 Cycles, For
200/115V 3 Ph 400 CPS 4 Wire System
Type D-1
Specification MIL-T-5383
Mfr's. Part No. _____
Mfr's. Serial No. _____
Order No. _____
Stock No. _____
Date of Manufacture _____
Quantity _____
(Mfr's. Name or Trademark)
Name of Contractor (if not the same as manufacturer)

6. NOTES

6.1 Use.- The Type D-1 transformer covered by this specification is intended for use in connection with metering, reactive current division, and governor control in 200/115V, 400 cycle ac systems for aircraft.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.3 Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on Qualified Products List AFQPL-5383. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Air Materiel Command and arrange to have the products that they propose to offer to the Air Force tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio, Attention: MCREXE.

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6.4 Sources.- When requesting publications, refer to both title and number.

6.4.1 Copies of American Standards Association Standard C57.10-1948 may be obtained upon application to the American Standards Association, 29 West Thirty-ninth Street, New York 18, New York.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

(Copies of this specification may be obtained from the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio.)

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MCREXE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1426 APR 63 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE PLATE NO. 15419

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PLATE NO. 15419 (BACK)

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